

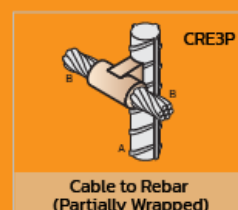
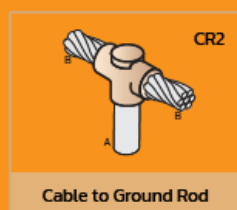
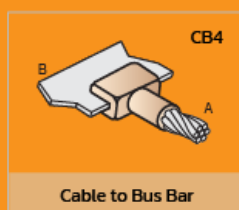
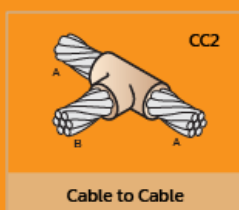


EXOTHERMIC WELDING

Exothermic Welding (a.k.a Thermite or Aluminothermic Welding] is a permanent molecular bonding process in which a chemical reaction between copper oxide and aluminum generates superheat with molten metals reaching high temperatures exceeding 2,600°C (4,000°F).

The process creates a homogeneous, pore-free connection with superior electrical conductivity, exceptional mechanical strength, and long-term corrosion resistance. Suitable for direct burial in soil or concrete, exothermic welds provide a maintenance-free, lifetime connection for critical grounding and bonding applications.

EXAMPLES OF EXOTHERMIC WELDED CONNECTION TYPES



GRAPHITE MOULD



- Kumwell Graphite Moulds are manufactured from high-quality graphite blocks, precision engineered to withstand extreme exothermic temperatures without degrading rapidly.
- Lifespan/Durability: Designed for a minimum of 50 welding operations under normal usage and storage conditions.

WELD METAL POWDER



Kumwell weld metal powder uses a specially formulated mixture of metallic powders, primarily copper oxide and aluminum. When ignited, these composite chemicals trigger a high-temperature exothermic reaction, forming a permanent, highly conductive copper alloy joint (typically over 93% copper) Tested & inspected in compliance with global standards UL467, IEEE837, IEC62561-1

- Nontoxic and heavy metal.
- Steady burn without slag, porosity pop or fire out.
- Consistency of color.

HANDLE CLAMPS



Kumwell Handle Clamps & Support Handles are used for handling / clamping the moulds in vertical position and tightening around the conductor / rebar / ground rod / rail tracks / steel structure etc., so as to avoid leakage during the Exothermic welding process.



TOOLKIT



Kumwell Standard Tools are specially designed to support the **Exothermic welding process**, ensuring efficiency, safety, and reliable performance.

- **Flint Ignitor:** for starting the reaction
- **Cable Clean Brush:** for cleaning round conductors and cables
- **Mould Brush:** for cleaning graphite moulds
- **Busbar Brush:** for cleaning flat conductors and busbars
- **Scraper:** for removes slag from the mould after welding

With Kumwell Standard Tools, every weld is performed with **greater precision, cleanliness, and long-term reliability**

Kumwell



***Not Include Tools**